Artificial Intelligence Evolution: From Predictive Modelling to Generative Content

Executive Summary

The technological landscape of Artificial Intelligence (AI) has evolved from basic machine-led mimicry of human behaviour to sophisticated systems capable of generating original content. This briefing explores the hierarchical relationship between Machine Learning (ML), Deep Learning (DL), and Generative AI (Gen AI). Central to this evolution is the scale of data—ranging from thousands of records to trillions of parameters—and the transition from predictive models, which forecast outcomes like health risks or market prices, to generative models that produce text, images, and code. Key findings include the distinction between supervised and unsupervised learning, the structural complexity of neural networks in Deep Learning, and the massive scale of Large Language Models (LLMs) such as GPT-5.

---

1. The Hierarchy of Artificial Intelligence

Artificial Intelligence is defined as a machine's ability to mimic human behavior. Within this broad field, specialized subsets have emerged based on the complexity of the data and the architecture of the models used.

- **Machine Learning (ML):** A subset of AI focused on predictive outcomes using datasets ranging from 1,000 to millions of records.
- **Deep Learning (DL):** A subset of ML that utilizes neural networks and handles vast amounts of data, typically in the millions to trillions of records.
- **Generative AI (Gen AI):** A specialized subset of Deep Learning that shifts the focus from prediction to the generation of new content (text, audio, video, or images).

---

2. Machine Learning: Predictive Methodologies

Machine Learning serves as the foundation for "Predictive AI." It relies on identifying patterns in historical data to forecast future events or categorize information.

Types of Machine Learning

- **Supervised Learning (Labelled Data):** This method uses datasets where the input is already tagged with the correct answer.
    - **Linear Regression:** Produces numerical or quantitative outputs. Examples include predicting home purchase prices in real estate or defect rates in manufacturing.
    - **Classification:** Groups data into specific categories or labels. A primary use case is medical diagnostics, such as determining if a patient has a risk of a heart attack (Yes/No).
    - **Content-Based Filtering:** Utilizes explicit user feedback, such as star ratings, likes, or dislikes, to make recommendations.
- **Unsupervised Learning (Unlabelled Data):** This method identifies patterns based on behaviour or genre without explicit labels.

    - **Collaborative Filtering:** Analyses user behaviour to find similarities without relying on direct comments or likes.

Use Case: Heart Disease Prediction via Retinal Imaging

The source context highlights the use of Convolutional Neural Networks (CNN) to analyse retinal images. By processing millions of records containing eye details and genetic history, the model can predict whether a new, unseen retinal image indicates a risk of heart disease.

---

3. Deep Learning and Neural Networks

Deep Learning differentiates itself from standard Machine Learning through the use of Neural Networks and the sheer volume of data processed. While ML models handle millions of records, DL models often scale to trillions.

- **Predictive Nature:** Despite its increased complexity, Deep Learning remains primarily a predictive tool.
- **Architecture:** It combines vast datasets with multi-layered neural networks to achieve higher accuracy in complex tasks, such as multimodal processing (accepting text, image, and video simultaneously).

---

4. Generative AI and Large Language Models (LLMs)

Generative AI represents a shift from predictive outputs to generative outputs. Unlike ML or DL, which might predict the next value in a sequence, Gen AI creates entirely new data.

Large Language Model (LLM) Capabilities

- **Functionality:** Gen AI can generate content across all domains, including software code, text, images, and audio.
- **The GPT Architecture:** Generative Pretrained Transformers (GPT) are pretrained on massive, cross-disciplinary datasets.
- **Scaling Complexity:** The evolution of these models is marked by an increase in layers and parameters:
    - **GPT-4:** Utilizes between 96 and 120 layers.
    - **GPT-5:** Features 190 layers and 120 billion parameters.

---

5. Data Management and Model Development

The efficacy of any AI model—whether predictive or generative—is dependent on the quality of the data and the rigor of the development process.

The Development Lifecycle

1. **Data Collection:** Gathering the necessary volume of records (millions to billions).
2. **Data Cleansing:** Ensuring the data is accurate and usable.

3. **Model Selection:** Choosing the appropriate architecture (e.g., CNN for images).
4. **Training and Testing:** Models are typically developed using a split-data approach:
    - **80% Training Data:** Used to teach the model patterns.
    - **20% Testing and Validation:** Used to verify the model's accuracy using data it has not seen during the training phase.

Variables in Predictive Modelling:

Models utilize specific parameters (columns) to generate predictions. In a medical context, these include:

- Demographics (Name, Age, Location)
- Biological markers (Blood group, Blood pressure, Diabetic status)
- Behavioural and Genetic factors (Genetic history, Smoker/Drinker status)